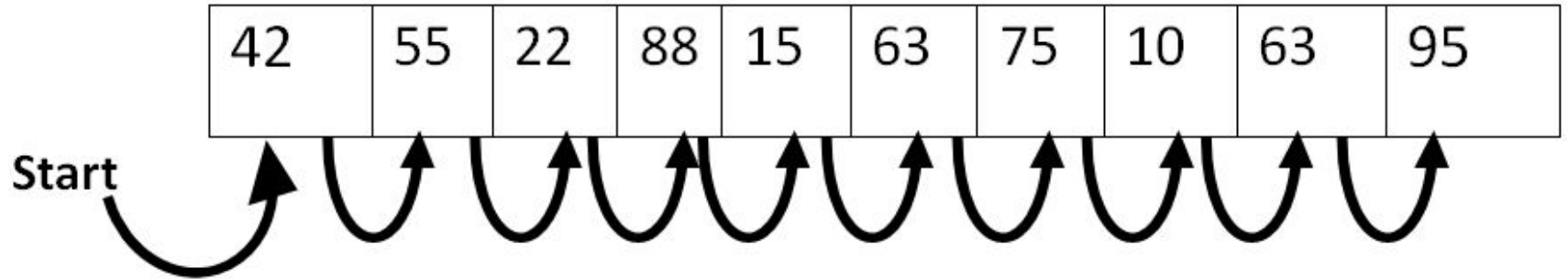

Data Structures

Hashing

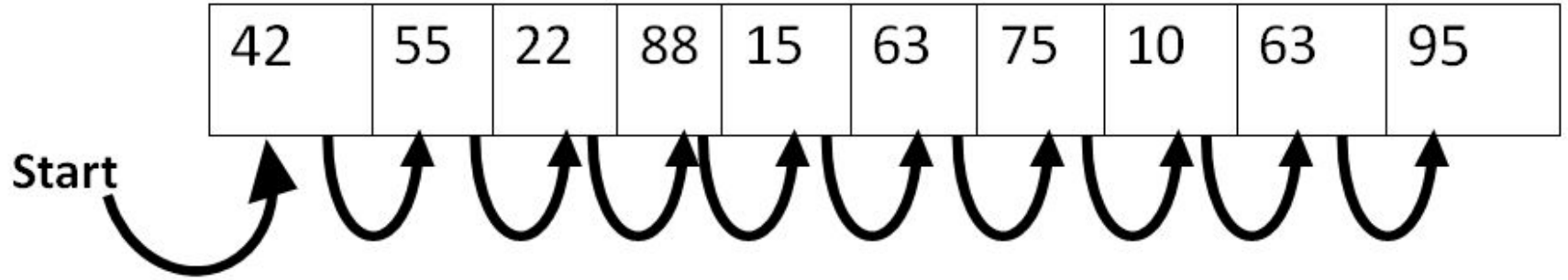
Slides taken from Prantik Paul [PKP]

Linear Search



Problem?

Linear Insert/Delete



Problem?

Linear Search

**What if it's Binary Search?
On a sorted array?**

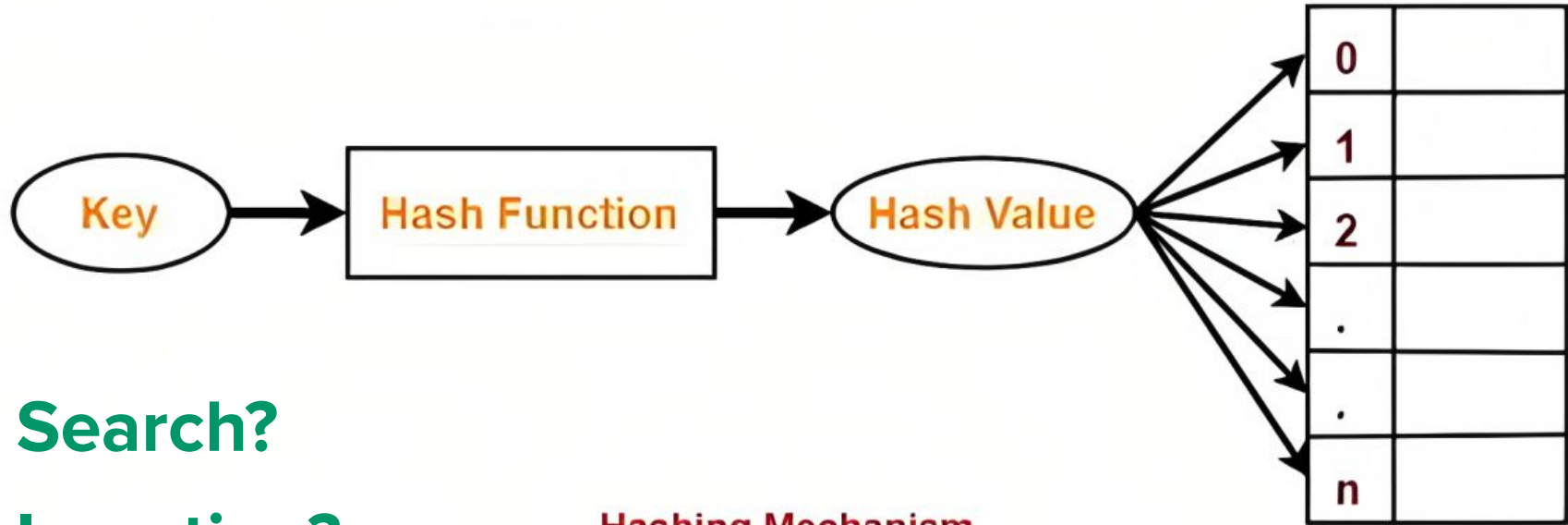
Linear Insert/Delete

What if it's on a sorted array?

Search/Insert/Delete

What if it's on a Linked List?

Solution? - Hashing



Hashing Mechanism

Search?

Insertion?

Deletion?

Key-Value Pairs

Code	Output
<pre>stu_id = {'Harry':12, 'Ron':15, 'Hermione':1} value = stu_id['Hermione'] print(value) print(stu_id['Harry'])</pre>	<pre>1 12</pre>

Key-Value Pairs

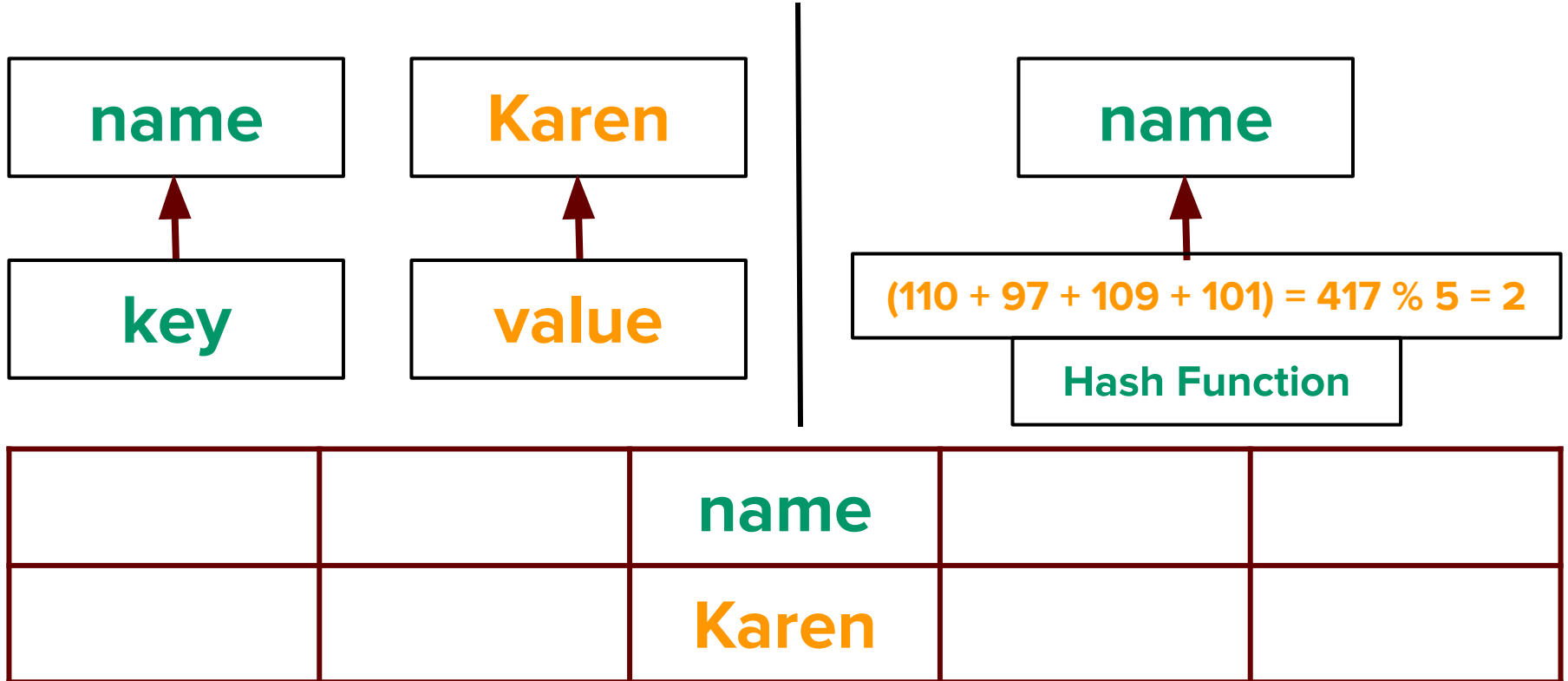
0	1	2	3	4
9341049	8828328	7100090	9889849	9651423

0	1	2	3	4
Mumit	Belal	Mukul	Muzil	Muhit

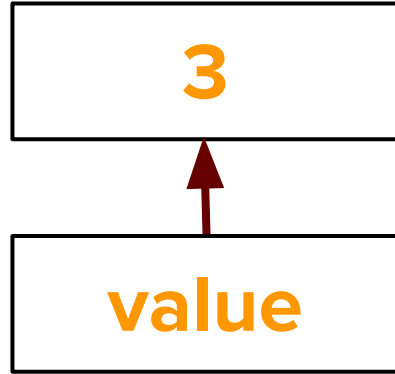
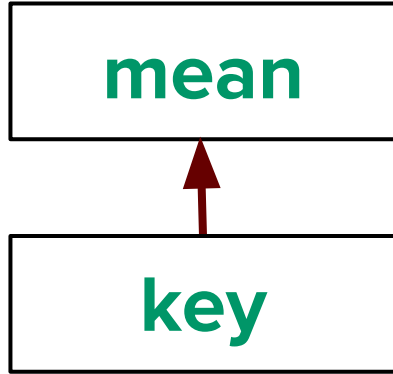
Key-Value Pairs

0	1	 700000	800000....	9889849
null	null	Mr. X	Ms. Y	Muzil

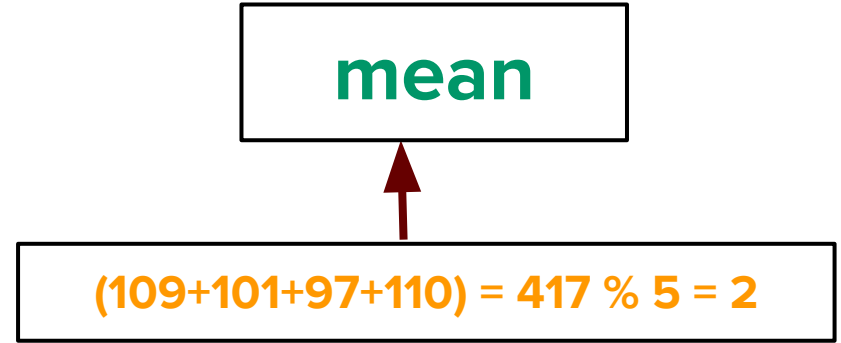
Hashing



Hashing



COLLISION!!



		name		
		Karen		

Hashing

		name		
		Karen		

mean
3

Hashing

		mean		
		3		



name
Karen

mane
no

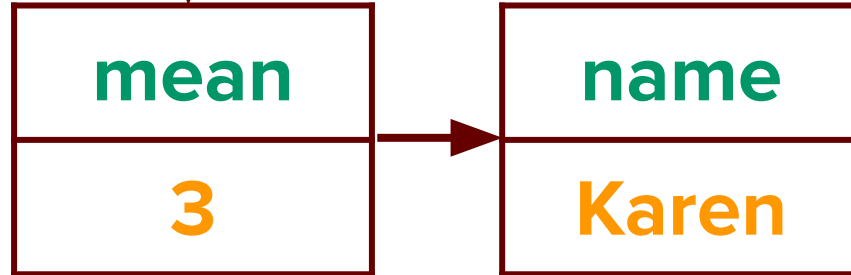
**Forward
Chaining**

Hashing

		mane		
		no		

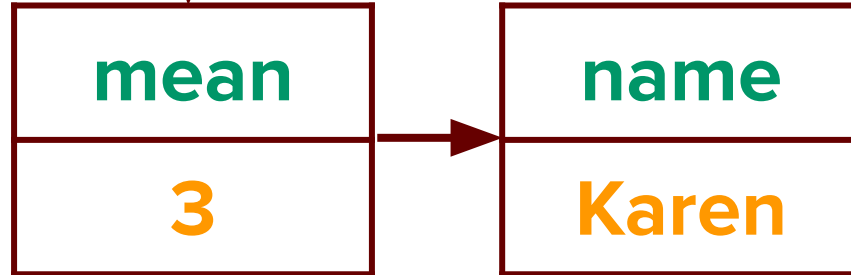


**Forward
Chaining**



Hashing

		mane		
		no		



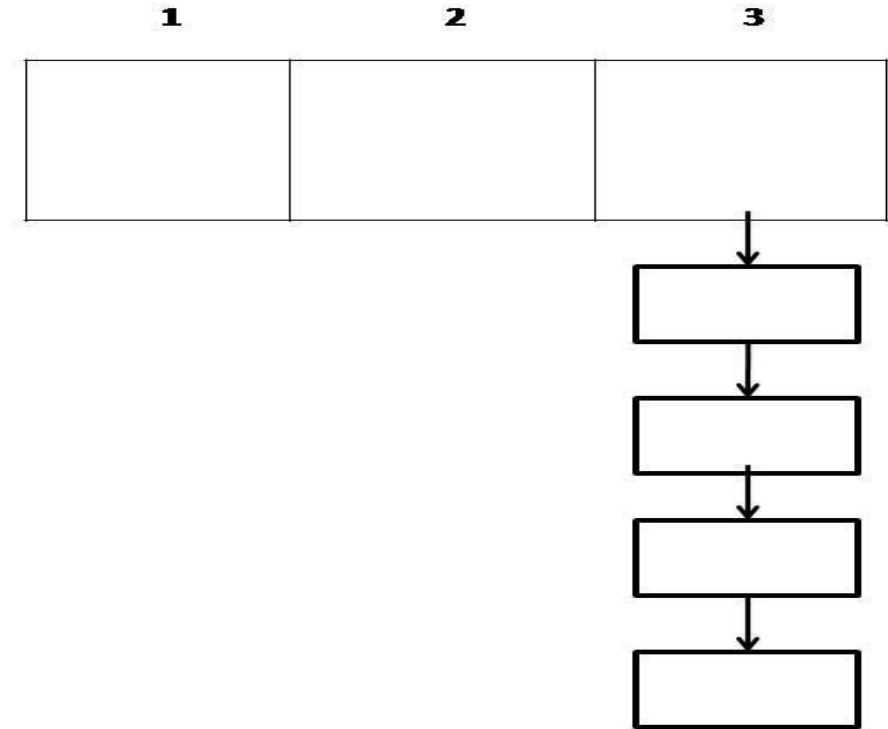
Worst Case?
Search/Insert/Delete?

Forward Chaining

How big should the array be?

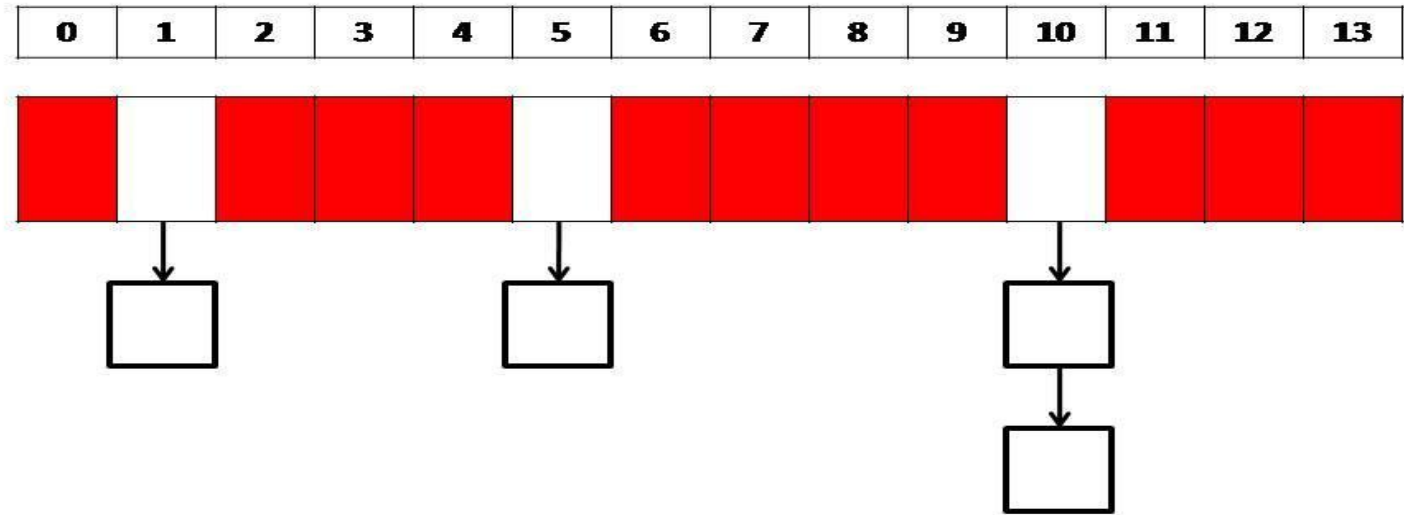
Forward Chaining

Too small



Forward Chaining

Too big



Forward Chaining (Insert)

```
def Forward_Chaining_Insert(arr):  
    hashtable = [None] * size  
  
    for element in arr:  
        hash_value = hash_func(elem)  
        if( hashtable[hash_value] == None):  
            hashtable[hash_value] = Node(elem, None)  
        else:  
            current = hash_table[hash_value]  
            hash_table[hash_value] = Node(elem, current)
```

Forward Chaining (Search)

```
def Forward_Chaining_Search(elem):  
    hash_value = hash_func(elem)  
    temp = hash_table[hash_value]  
  
    while (temp!=None):  
        if(temp.elem == elem):  
            return True  
        temp = temp.next  
  
    return False
```

Forward Chaining

$$\text{hash}(\text{key}) = \text{key} \% 5$$

```
Insert (Key: 12 Value: "Apple")
Insert (Key: 5 Value: "Orange")
Insert (Key: 17 Value: "Banana")
Insert (Key: 10 Value: "Grapes")
Insert (Key: 22 Value: "Watermelon")
Insert (Key: 15 Value: "Pineapple")
```

Hashing

		name	mean	
		Karen	3	

Linear
Probing

Hashing

		name		
		Karen		

$\text{hash1}(\text{mean}) = 2$

$\text{hash2}(\text{mean}) = 7$

$\text{Location} = (2 + i \times 7) \% 5$

Double Hashing

Double Hashing

KEYS : 79, 69, 98, 72, 14, 50

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	

Hash table

$$h_1(k) = k \bmod 13$$

$$h_2(k) = 1 + (k \bmod 11)$$

$$\text{Index for } k = (h_1(k) + i \times h_2(k)) \% 13$$